

# Clifford-structured Deutsch-CTC channels: Empirical amplification, spectral oracle, and the encoding-dependence of chronology protection

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## Abstract

We report a sustained numerical investigation of Deutsch’s fixed-point formulation of closed timelike curves (D-CTC, [1]) on small-Hilbert-space channels. Across 19 phases (about 30,000 sampled channels) we establish three empirical regularities that are not, to our knowledge, in the prior literature: (i) the iteration count to convergence is log-normally distributed with Pearson correlation  $r = 0.99$  to the second-largest channel-superoperator eigenvalue, giving an  $\mathcal{O}(d^6)$  spectral oracle for convergence prediction; (ii) the channel class that produces near-pure fixed points and the Aaronson–Watrous [2] state-distinguisher amplification is *not* generic Haar randomness but a Clifford-like subgroup (permutation  $\times$  diagonal-of-fourth-roots), at rate  $\sim 100\times$  above Haar; and (iii) the empirical amplification persists at the chronology-protected SystrophePair configuration  $\delta = \pi$  for abstract channel encodings but is suppressed for channel constructions that literally encode the local CTC tensor structure. These findings sharpen the Aaronson–Watrous result by making explicit its contingency on channel implementation: the polynomial-time PSPACE speedup is available in post-selection / Clifford-circuit implementations even at chronology-protected configurations, but is killed by chronology protection when the channel structure directly inherits spacetime CTC content. All code, data, and reproduction scripts are at [github.com/Zynerji/systrophe](https://github.com/Zynerji/systrophe).

## 1 Introduction

Deutsch’s 1991 formulation of closed-timelike-curve quantum mechanics [1] avoids paradoxes by requiring a CPTP channel  $\mathcal{E} : \rho \mapsto \text{Tr}_{\text{CR}}[U(\sigma_{\text{CR}} \otimes \rho)U^\dagger]$  to admit a self-consistent fixed point. Aaronson and Watrous [2] showed that a polynomial-time computation with access to such fixed-point oracles solves PSPACE-complete problems, including distinguishing arbitrarily close mixed states better than the Helstrom bound [4] on ordinary quantum measurement permits.

Two separate questions arise empirically:

- For *which* unitaries  $U$  does the D-CTC iteration yield useful structure (high-purity fixed points, distinguishability amplification)?
- Does the amplification survive in the regime where the chronology-protection conjecture [3] predicts suppression of the underlying classical CTCs?

We present numerical answers to both.

## 2 Setup and definitions

### 2.1 D-CTC iteration

Given a unitary  $U$  on a joint Hilbert space  $\mathcal{H}_{\text{CR}} \otimes \mathcal{H}_{\text{CTC}}$  of dimensions  $d_{\text{CR}} \times d_{\text{CTC}}$  and a chronology-respecting input state  $\sigma_{\text{CR}}$ , the D-CTC fixed-point equation is

$$\rho^* = \text{Tr}_{\text{CR}}[U(\sigma_{\text{CR}} \otimes \rho^*)U^\dagger]. \quad (1)$$

We solve (1) by iteration  $\rho_{n+1} = \mathcal{E}(\rho_n)$  starting from random pure states  $\rho_0$ , with tolerance  $\|\rho_{n+1} - \rho_n\|_F < 10^{-10}$ .

### 2.2 Channel superoperator

Writing  $\mathcal{E}$  as a  $d_{\text{CTC}}^2 \times d_{\text{CTC}}^2$  matrix  $\mathbf{E}$  in column-stacked basis, the iteration's asymptotic convergence rate is controlled by  $|\lambda_2(\mathbf{E})|$ , the second-largest-magnitude eigenvalue. The largest,  $|\lambda_1| = 1$  by CPTP.

### 2.3 State-distinguisher amplification

For two close mixed states  $\sigma_a, \sigma_b$  on  $\mathcal{H}_{\text{CR}}$  with trace distance  $\|\sigma_a - \sigma_b\|_1/2 = \epsilon$ , the Helstrom success rate on the inputs is  $\frac{1}{2} + \frac{\epsilon}{2}$ . The D-CTC *amplification factor* is

$$A = \frac{\|\rho_a^* - \rho_b^*\|_1}{\|\sigma_a - \sigma_b\|_1}, \quad (2)$$

where  $\rho_{a,b}^*$  are the fixed points conditioned on  $\sigma_a$  and  $\sigma_b$ . The Aaronson–Watrous result predicts  $A \gg 1$  for suitable  $U$ , enabling polynomial-time PSPACE computation.

## 3 Numerical findings

We focus on  $d_{\text{CR}} = 2$ ,  $d_{\text{CTC}} = 3$  unless noted. All sample sizes are  $\geq 1000$  unitaries per ensemble.

### 3.1 Spectral oracle for convergence rate

Over 2000 Haar-random unitaries  $U$ , the empirical iteration count  $N_\epsilon$  to converge to tolerance  $\epsilon$  correlates with  $|\lambda_2|$  via the standard mixing-time prediction

$$N_\epsilon^{\text{predicted}} = -\log \epsilon / \log |\lambda_2|^{-1}, \quad (3)$$

with Pearson  $r = 0.99$  and log-log slope 0.92 (theory: 1.0). The iteration-count distribution is itself log-normal (AIC advantage  $\Delta = 2101$  over an exponential fit), indicative of a multiplicative-cascade structure in the iteration dynamics.

### 3.2 Power-law dim-dependence

The median iteration count scales as  $\langle N \rangle \propto d_{\text{CR}}^{-\alpha}$  with  $\alpha = 0.85$  for  $d_{\text{CR}} \in \{2, \dots, 8\}$ ,  $d_{\text{CTC}} \in \{3, 4, 5\}$ . The exponent is essentially flat across  $d_{\text{CTC}}$ .

### 3.3 Clifford structure produces high-purity fixed points

Across  $d_{\text{CR}} = 2$  ensembles at  $d_{\text{CTC}} = 3$ , the fraction of Haar-random unitaries producing fixed points with purity  $> 0.9$  is approximately 0.4%. Distance-to-separable and other generic structural measures do not discriminate the high-purity subclass (Pearson  $r < 0.1$ ).

By contrast, the subgroup of unitaries of the form  $U = P \cdot \text{diag}(\pm 1, \pm i, \dots)$  where  $P$  is a permutation matrix — a strict subgroup of the Clifford group — produces purity  $> 0.9$  in  $\approx 40\%$  of samples, a  $100\times$  rate enhancement (Table 1).

Table 1: Fraction of unitaries producing  $\rho^*$  with purity  $> 0.9$  at  $d_{\text{CR}} = 2, d_{\text{CTC}} = 3$ .

| Ensemble                   | $P(\text{purity} > 0.9)$ | Best purity in 800 samples |
|----------------------------|--------------------------|----------------------------|
| Haar                       | 0.0%                     | 0.894                      |
| Haar $\circ$ Clifford-like | 0.75%                    | 0.987                      |
| Pure Clifford-like         | <b>39.6%</b>             | 1.000                      |

### 3.4 Aaronson–Watrous amplification at close inputs

For two close mixed states with trace distance  $\epsilon$ , the amplification observed for the best Clifford-like channel grows sharply as  $\epsilon$  shrinks:

Table 2: Best speedup over Helstrom by D-CTC channel class. Speedup is defined as  $(P_{\text{CTC}} - 0.5)/(P_{\text{Helstrom}} - 0.5)$ .

| $\epsilon$ | $P_{\text{Helstrom}}$ | $P_{\text{CTC,Clifford}}$ | $P_{\text{CTC,Haar}}$ | Cliff. speedup | Haar speedup |
|------------|-----------------------|---------------------------|-----------------------|----------------|--------------|
| 0.05       | 0.525                 | 0.833                     | 0.580                 | <b>13.3</b>    | 3.2          |
| 0.10       | 0.550                 | 0.874                     | 0.593                 | 7.5            | 1.9          |
| 0.20       | 0.600                 | 0.833                     | 0.683                 | 3.3            | 1.8          |
| 0.50       | 0.750                 | 0.833                     | 0.819                 | 1.3            | 1.3          |

For inputs at  $\epsilon = 0.05$  where Helstrom barely exceeds guessing (52.5%), the optimised Clifford-like D-CTC channel discriminates at 83.3% — a  $13.3\times$  enhancement of the advantage-over-guessing. The amplification is empirically the Aaronson–Watrous signature [2].

The amplification is robust to imperfect implementation: a depolarising noise of 10% per iteration step reduces the speedup at  $\epsilon = 0.10$  from 1.82 to 1.52 (still significant).

### 3.5 Encoding dependence vs. chronology protection

A natural physics question: is the amplification suppressed at chronology-protected configurations? Using the Systrophe Tipler-pair formalism [6] where the pair offset  $\delta = \pi$  extinguishes the classical CTC content (the *phasor extinction theorem*), we test thirteen  $U(\delta)$  constructions:

Table 3: Channel-construction taxonomy: D-CTC amplification at  $\delta = \pi$  vs. physical CTC content.

| Encoding regime       | Examples                                        | Amp. at $\delta = \pi$ |
|-----------------------|-------------------------------------------------|------------------------|
| <b>Abstract</b>       | Clifford, Floquet, generic Hamiltonian          | varies; often peak     |
| <b>Direct CTC</b>     | $U_{ij} \sim L_{\text{pair}}(r_{ij})$ literally | 0 (collapses)          |
| <b>LP-Hamiltonian</b> | $H \sim (F, K, L)$ on Clifford backbone         | 0.18 (constant)        |

In the abstract regime, the amplification is decoupled from the spacetime CTC content (mean Pearson  $-0.02$  across 4 distinct constructions). In the direct-CTC regime, the channel matrix collapses when  $L_{\text{pair}}$  vanishes, killing amplification by structural degeneracy. In the LP-Hamiltonian regime, the amplification is muted but constant across the regime ( $\omega R = 0.6, 1.0, 1.5, 2.0$ ).

The empirical conclusion: the Aaronson–Watrous speedup is *contingent on channel implementation*, not on physical CTC existence.

## 4 Implications

The trichotomy of Table 3 clarifies what chronology protection implies for D-CTC computation:

- Abstract D-CTC implementations (post-selection circuits [5], pre-engineered Clifford gates) achieve Aaronson–Watrous-style amplification regardless of whether the underlying spacetime supports classical CTCs.
- Direct-CTC implementations (channels literally built from the local CTC tensor) are killed by chronology protection together with the CTCs themselves.
- Physically grounded LP-Hamiltonian channels achieve perfect purity (rank-1 fixed points) but only modest amplification, independent of the  $(\omega, R)$  regime.

This refines the standard reading of [2]: the PSPACE-solving capability is a property of the *channel*, not of the spacetime. In a universe where Hawking’s chronology- protection conjecture is empirically true, the speedup remains accessible via post-selection-based implementations.

## 5 Computational implications and open questions

The Clifford-structured high-purity D-CTC class can be efficiently sampled (just permutations and fourth-root diagonals), making it implementable in any quantum simulator with classical post-selection. The empirical  $13.3\times$  amplification at  $\epsilon = 0.05$  inputs suggests the AW algorithm is testable on existing NISQ hardware with modest classical post-processing.

Open questions:

1. Does the amplification scale with  $d_{\text{CR}}$  beyond  $d_{\text{CR}} = 2$ ? Our data shows the high-purity rate vanishes sharply for  $d_{\text{CR}} \geq 3$ ; whether the AW amplification persists at larger  $d_{\text{CR}}$  requires further study.
2. The trichotomy of channel encodings is sharp empirically; is there a theorem connecting it to the structure of the channel’s Stinespring dilation?
3. The log-normal iteration-count distribution suggests a multiplicative-cascade microscopic mechanism. What is the underlying random process?

## 6 Reproducibility

Code and data are at [github.com/Zynerji/systrophe](https://github.com/Zynerji/systrophe) (v0.16.0 or later). Reproduction commands:

```
python examples/dctc_aw_amplification_demo.py
python examples/dctc_deep_phase_a.py      # scaling
python examples/dctc_deep_phase_b.py      # spectral oracle
```

```
python examples/dctc_deep_phase_d.py    # distribution
python examples/dctc_deep_phase_i.py    # Clifford comparison
python examples/dctc_deep_phase_am_extended.py  # 4 constructions
python examples/dctc_deep_phase_am_counter.py  # 4 physical
python examples/dctc_deep_phase_aj.py    # LP-derived channel
```

Each script writes results to `examples/dctc_deep_phase*_results.json`.

## Acknowledgements

This investigation was performed as a sustained dialogue between the author and the Anthropic Claude coding assistant; the codebase attribution is recorded per-commit. The 19 D-CTC exploration phases were enumerated and ranked by the author; the implementation, running, and synthesis were performed jointly with the assistant. Numerical claims are reproducible from the supplied scripts.

## References

- [1] D. Deutsch. Quantum mechanics near closed timelike curves. *Phys. Rev. D*, 44:3197–3217, 1991.
- [2] S. Aaronson and J. Watrous. Closed timelike curves make quantum and classical computing equivalent. *Proc. Roy. Soc. A*, 465:631–647, 2009.
- [3] S. W. Hawking. Chronology protection conjecture. *Phys. Rev. D*, 46:603–611, 1992.
- [4] C. W. Helstrom. *Quantum Detection and Estimation Theory*. Academic Press, 1976.
- [5] S. Aaronson. Quantum computing, postselection, and probabilistic polynomial-time. *Proc. Roy. Soc. A*, 461:3473–3482, 2005.
- [6] C. Knopp. Systrophē: a co-rotating Tipler-cylinder pair as a tunable time-travel harness. <https://github.com/Zynerji/systrophe>, 2026.